PHYSICAL CHEMISTRY CUMULATIVE EXAM

March 28, 1992

Rotational Spectra

The principal moments of inertia, A, B, and C, for a molecule containing n atoms are the solutions for I of the determinantal equation

$$\begin{vmatrix} I_{xx}-I & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy}-I & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz}-I \end{vmatrix} = 0$$

when by convention $C \geq B \geq A$ and

$$I_{xx} = \sum_{i=1}^{n} m_i(y_i^2 + z_i^2)$$

and

$$I_{xy} = \sum_{i=1}^{n} m_i x_i y_i$$

(a) Determine the principal moments of inertia for CH_4 if the center of mass is located at the carbon atom. In this system the coordinates of the atoms are:

C (0, 0, 0)

H_1 (0, 0, 1.091)

H_2 (1.028, 0, -0.364)

H_3 (-0.514, 0.890, -0.364)

H_4 (-0.514, -0.890, -0.364)

where $r_{C\text{-}H}$ = 1.091Å. The atomic mass unit is 1.66054×10^{-27} kg and $m_p = 1.67262 \times 10^{-27}$ kg, $m_N = 1.67493 \times 10^{-27}$ kg and $m_e = 9.10939 \times 10^{-31}$ kg.

(b) Molecules are divided into 3 general classes: (1) spherical tops where A = B = C, (2) symmetrical tops where A < B = C (prolate) or A = B< C (oblate) and (3) asymmetrical tops where A < B < C.

What is the classification of CH_4?

(c) A linear molecule is a special case of a prolate top with A = 0 and B = C. Such a molecule will have a rotational spectrum only if what condition exists?